

GIBRALTAR AP, GibraltarWMO: **084950**Lat: **36.1521N**Lon: **5.3494W**Elev: **5**StdP: **101.27**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.1	-1.9	3.2	12.4	0.8	4.0	12.3	18.1	14.0	15.6	14.0	3.7	270	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	30.8	20.3	29.1	20.0	27.9	20.1	23.5	25.9	23.0	25.5	22.5	25.1	6.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.4	24.9	22.1	16.8	24.5	21.6	16.3	24.1	69.8	26.0	67.9	25.5	66.0	25.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.2	10.7	DB	5.7	34.4	1.6	2.0	4.5	35.8	3.5	37.0	2.6	38.1	1.4	39.6	
			WB	2.5	24.8	1.7	1.2	1.3	25.7	0.3	26.3	-0.7	27.0	-2.0	27.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	13.7	14.0	15.4	16.9	19.1	22.0	24.1	24.6	22.7	20.1	16.6	14.6
	DBStd	4.22	1.82	1.83	1.69	1.68	1.85	1.72	1.82	1.61	1.79	1.81	2.19	1.80
	HDD10.0	3	2	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	596	143	120	91	49	12	0	0	0	0	6	61	114
	CDD10.0	3171	116	114	168	206	284	359	438	453	380	312	197	144
	CDD18.3	723	0	0	1	5	37	109	180	194	130	59	7	0
	CDH23.3	3069	0	0	3	7	74	358	1011	1131	413	67	5	0
(14) Wind	WSAvg	5.4	5.7	6.2	6.1	5.7	5.2	5.2	5.0	4.9	4.8	4.6	5.3	5.9
	PrecAvg	753	113	102	82	62	30	7	1	1	22	75	125	146
	PrecMax	1888	507	485	313	173	118	53	8	10	129	204	479	652
	PrecMin	338	0	0	0	0	0	0	0	0	0	0	1	0
	PrecStd	311	110	86	65	42	28	10	2	2	31	62	103	148
	DB	19.2	20.6	23.2	24.2	27.9	30.1	34.0	32.9	30.0	27.0	23.7	20.2	17.2
	MCWB	14.2	14.6	15.4	16.4	18.3	19.6	20.6	21.0	20.4	20.1	17.1	15.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	18.1	18.9	20.6	22.2	25.2	28.2	31.0	30.9	28.1	25.1	21.8	19.0	
	MCWB	13.9	13.8	14.3	15.5	17.6	19.5	20.3	20.4	19.8	19.1	16.8	15.1	
	DB	17.2	17.7	19.1	20.9	23.6	26.7	29.1	29.2	26.8	23.7	20.4	18.0	
	MCWB	13.6	13.7	14.0	15.2	16.9	19.0	20.1	20.3	20.0	18.9	16.5	14.8	
	DB	16.3	16.8	18.1	19.8	22.2	25.2	27.6	27.9	25.5	22.7	19.5	17.2	
	MCWB	13.3	13.4	13.9	14.9	16.6	19.0	20.1	20.7	20.3	18.8	16.4	14.6	
	WB	16.1	16.0	16.7	18.2	20.1	22.1	23.9	24.3	23.5	21.7	19.2	17.2	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	17.8	18.3	19.8	21.6	24.3	25.5	26.5	26.7	25.5	23.9	21.0	18.9	
	WB	15.1	15.2	15.8	16.9	19.1	21.2	23.1	23.6	22.7	20.7	18.3	16.2	
	MCDB	16.9	17.3	18.4	20.0	22.6	24.7	25.9	26.1	24.9	22.8	20.2	17.8	
	WB	14.5	14.6	15.2	16.2	18.3	20.7	22.5	23.2	22.2	20.2	17.6	15.6	
	MCDB	16.3	16.7	17.7	19.1	21.6	24.1	25.4	25.7	24.4	22.3	19.7	17.2	
	WB	13.9	14.0	14.7	15.7	17.7	20.2	22.0	22.7	21.6	19.6	16.9	15.0	
	MCDB	15.9	16.0	17.2	18.5	20.8	23.5	25.1	25.3	23.9	21.7	19.0	16.8	
(35) Mean Daily Temperature Range	MDBR	4.8	4.9	5.1	5.5	5.8	6.3	6.9	6.5	5.5	4.7	4.6	4.3	
	5% DB	5.8	6.4	7.0	7.3	8.0	8.6	9.6	9.2	7.6	6.3	5.6	5.2	
	MCWBR	3.8	3.7	3.7	3.6	3.6	3.6	4.0	3.9	3.2	3.0	3.2	3.3	
	5% WB	4.6	5.1	5.6	5.6	6.3	6.5	6.7	5.8	4.8	4.6	4.4	4.0	
	MCWBR	3.6	3.5	3.5	3.5	3.5	3.4	3.8	3.4	2.7	2.7	2.9	3.1	
(40) Clear-Sky Solar Irradiance	taub	0.336	0.361	0.402	0.400	0.416	0.437	0.438	0.450	0.426	0.397	0.360	0.341	
	taud	2.511	2.439	2.308	2.340	2.323	2.271	2.263	2.235	2.333	2.416	2.488	2.511	
	Ebn at Noon	859	875	864	884	871	848	844	827	829	822	821	827	
	Edh at Noon	82	98	122	124	128	135	135	136	118	100	83	77	
(44) All-Sky Solar Radiation	RadAvg	2.65	3.43	4.55	5.67	6.77	7.74	7.84	6.90	5.34	3.87	2.82	2.33	
	RadStd	0.25	0.42	0.49	0.41	0.49	0.27	0.17	0.25	0.30	0.36	0.24	0.24	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62

Nomenclature: See separate page